

HW1

R1. What is the difference between a host and an end system? List several different types of end systems. Is a Web server an end system?

There is no difference between a host and an end system, They're used interchangeably. Though a host typically refers to a service/app being offered, while an end system typically refers to a device, it just depends on the context.

Types of end systems:
PC, Laptop, Smartphone, consoles, cameras, etc.

Yes, Web servers are end systems.

R3. Why are standards important for protocols?

"it's important that everyone agrees on what each and every protocol does, so that people can create systems and products that interoperate."

In other words, it's important to have standards for protocols so developers can build products that are compatible and can communicate with systems made by others.

R4. List four access technologies. Classify each one as home access, enterprise access, or wide-area wireless access.

DSL: Home access
Cable: Home access
Ethernet: Enterprise access
5G: wide-area wireless access

R5. Is HFC transmission rate dedicated or shared among users?

HFC is shared among users. The text discusses if several users are downloading a file at once the rate will be slower. I can imagine this is what happened when we all tried to check our speed during class.

R7. What is the transmission rate of Ethernet LANs?

Range from 10 Mbps to 100 Mbps, 1 Gbps and 10 Gbps

R12. What advantage does a circuit-switched network have over a packet-switched network? What advantages does TDM have over FDM in a circuit-switched network?

A circuit-switched network will guarantee a constant fixed bandwidth for the duration of the call without congestion delays which is something most packet-switched networks cannot guarantee. In TDM, each circuit gets all of the bandwidth for a time slot while in FDM the bandwidth of a link is divided up among the connections established across the link. The advantage TDM has is that it does not require sophisticated analog hardware to shift signals into different frequency bands.

R15. Some content providers have created their own networks. Describe Google's network. What motivates content providers to create these networks?

Google built its own massive private network to connect its data centers globally, allowing them to bypass the public Internet for a significant amount of their traffic. The main reason they do this is to save money by reducing what they have to pay to ISPs, but it also gives them much better control over performance, ensuring their services stay fast and reliable for users.

R16. Consider sending a packet from a source host to a destination host over a fixed route. List the delay components in the end-to-end delay. Which of these delays are constant and which are variable?

The delay components are processing delays, transmission delays, propagation delays, and queuing delays. All of these delays are fixed, except for the queuing delays, which are variable.

R23. What are the five layers in the Internet protocol stack? What are the principal responsibilities of each of these layers?

Application Layer: Supporting network applications (HTTP, SMTP, FTP, DNS).

Transport Layer: Providing host-to-host data transfer (TCP, UDP).

Network Layer: Routing of datagrams from source to destination (IP).

Link Layer: Data transfer between neighboring network elements (Ethernet, WiFi, Cable).

Physical Layer: Moving the individual bits "on the wire".

R28. Suppose Alice and Bob are sending packets to each other over a computer network. Suppose Trudy positions herself in the network so that she can capture all the packets sent by Alice and send whatever she wants to Bob; she can also capture all the packets sent by Bob and send whatever she wants to Alice. List some of the malicious things Trudy can do from this position.

Since Alice and Bob are communicating through an unsecured network, Trudy is in the middle with the ability to sniff the packets being passed which may contain several forms of sensitive

data along with possibly placing a sniffer making a copy of every packet going to/from the network they're communicating on. Another malicious thing Trudy can do is IP spoof which involves sending packets using a false source address which is a way one user can masquerade as another user.